Raspberry Pi Pico

P1
rpi pico 1 W
2648-SC0918CT-ND
smd

+3.3V
220 R1
3V1 green
GND

R10
4.7k

+BATT
GND
+3.3V
5V
J1

NT5 NetTie_2
NT6 NetTie_2
NT7 NetTie_2
NT8 NetTie_2

D1
D_Schottky
+3.3V

R2
220
D13
blue
GND

bt1
bt2
bt3
bt4

3V3_EN
RUN
GP0
GP1
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
GP14
GP15
GP16
GP17
GP18
GP19
GP20
GP21
GP22
GP26
GP27
GP28
VBUS
VSYS
3V3_OUT
ADC_VREF
AGND
GND
GND_1
GND_2
GND_3
GND_4
GND_5
GND_6

sda
scl
right_2
right_1
right_en
left_2
left_en
left_1

J3
joystick
J4
speed knob

* these pins are 5V tolerant inputs.
The input is divided down from 5v to a safe level.
If you want to use this pin as a 3.3v input or as an output,
solder the NetTie pads on the back of the board.

left motor

U1
DRV8245HQPWPRQ1

LVL6, in independent mode that means faults are retried

SR
DIAG
PH/IN2
EN/IN1
DRVOFF
VM
OUT1
OUT2
GND
EPAD
ITRIP
MODE
NFAULT
IPROPI
NSLEEP

left_2
left_1
left_ipropi
left_en

C1
0.1uF

8.2k +-10% LVL2 INDEPENDENT
8.2k
R7

HSL1
Heatsink

J100
0197054001
J101
0197054001
optionally replace J7 to be weelye compatible

ML+
ML-
J7
1017491

right motor

U2
DRV8245HQPWPRQ1

right_1
right_2
right_ipropi
right_en

C2
0.1uF

8.2k +-10% LVL2 INDEPENDENT
8.2k
R6

HSR1
Heatsink

MR-
MR+
J8
1017491

optionally replace J8 to make the PCB weelye compatible
weelye style J104
weelye style J105
J106
0197054001
J107
0197054001

ADC (for measuring battery voltage and motor current)

U4
TLA2024IRUGR

+3.3V
NT1
NetTie_3
TP0
TestPoint
+BATT
R13
100k
R14
8.2k
right_ipropi
left_ipropi
ADDR
AIN0
AIN1
AIN2
AIN3
NC
VDD
SCL
SDA
scl
sda
C4
0.1uF
pullup resistors
TP1
TestPoint
left_ipropi
TP2
TestPoint
right_ipropi
R11
604
R12
604
R15
3.3k
R16
3.3k

qwiic I2C plug
J2
Conn_01x04
MP

weelye compatible header
J11
Conn_01x10_Pin
switch
bat-
D2
D_Schottky

3.5mm audio jacks for if you want to connect buttons as an alternative to the joystick
B1
ASJ-200-B-HT
B2
ASJ-200-B-HT
B3
ASJ-200-B-HT
B4
ASJ-200-B-HT

battery power circuit
J102
0197054001
J103
0197054001
optionally replace J5 to be weelye compatible
J5
TB005-762-02BE
screw terminal
SW1
switch
R3
3.3k
R4
22.1k
R5
22.1k
18V zener
D4
D3
PSMN1R0-40YLDX
Q1
on/off switch
PSMN1R0-40YLDX
Q2
reverse polarity protection
C5
33uf
J6

alternative voltage regulator
Instead of the voltage regulator circuit, you could get a single THT module that outputs 5V.
Make sure the pinout matches.
Make sure the min and max input voltage are compatible with your battery.
U5
D78B05T, RBT05W24S05, pololu5593
IN
OUT
GND

voltage regulator circuit https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=FA36C6A68A48DC1D
U3
LMR54410DBVR
CB
GND
FB
SW
VIN
EN
C6
0.1uF
C3
1uF
C7
0.1uF
L1
10uH
R8
115k
R9
22.1k
C8
22uF
5V

3mm diameter bolt holes
3.35in x 2.2in
85.09mm x 55.88mm
H1 M3
H2 M3
H3 M3
H4 M3

M1
M2
optional cooling fans
wire to J6
MF40200V1-1000U-A99 (5V, 3.5-6)
MF40201V1-1000U-A99 (12V, 4.5-13.8)
MF40202V1-1000U-A99 (24V, 8-27.6V)

FID1 Fiducial
FID2 Fiducial
FID3 Fiducial

https://github.com/gobabygocarswithjoysticks gobabygocarswithjoysticks@gmail.com
Sheet: /
File: gbg-pcb.kicad_sch
Title: GBG-PCB
Size: A3
Date:
Rev: v2.3
KiCad E.D.A. 10.0.6
Id: 1/1